Himanshu Rai

MRP1_Jyoti_Darshan_Project_Report

EMBA

Document Details

Submission ID

trn:oid:::3618:132397616

Submission Date

Mar 21, 2026, 2:49 PM GMT+5:30

Download Date

Mar 21, 2026, 2:58 PM GMT+5:30

File Name

MRP1_Jyoti_Darshan_Project_Report.docx

File Size

259.4 KB

26 Pages

9,697 Words

56,244 Characters

4% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Small Matches (less than 8 words)

Match Groups

18 Not Cited or Quoted 3%
Matches with neither in-text citation nor quotation marks

9 Missing Quotations 1%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

3% Internet sources
1% Publications
3% Submitted works (Student Papers)

Match Groups

- **18** Not Cited or Quoted 3%
  Matches with neither in-text citation nor quotation marks
- **9** Missing Quotations 1%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

3% Internet sources
1% Publications
3% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	caedes.net	1%
2	Internet	iafor.org	<1%
3	Internet	www.researchsquare.com	<1%
4	Internet	elibrary.tucl.edu.np	<1%
5	Student papers	SP Jain School of Global Management on 2026-03-05	<1%
6	Internet	jmhorizons.com	<1%
7	Internet	trepo.tuni.fi	<1%
8	Student papers	Birkbeck College on 2015-01-30	<1%
9	Internet	kclpure.kcl.ac.uk	<1%
10	Student papers	University of Leicester on 2026-01-23	<1%

11 Internet

intranet.xavient.com <1%

12 Internet

pmc.ncbi.nlm.nih.gov <1%

13 Internet

www.telegraphindia.com <1%

14 Publication

"Marketing and Smart Technologies", Springer Science and Business Media LLC, 2... <1%

15 Student papers

King's College on 2024-05-22 <1%

16 Student papers

University of Sunderland on 2007-11-28 <1%

17 Student papers

University of Surrey on 2012-11-25 <1%

18 Publication

Lecture Notes in Computer Science, 2015. <1%

19 Publication

Masa'deh, Ra'ed (Moh'd Taisir), Ali Tarhini, Ashraf Bany Mohammed, and Mahmo... <1%

20 Publication

Styles, Christopher. "The Rise of Mobile Malware: Challenges in Securing Mobile B... <1%

21 Student papers

University of Bath on 2015-03-19 <1%

22 Student papers

Walden University on 2010-10-23 <1%

23 Internet

olj.onlinelearningconsortium.org <1%

IIM RANCHI
बहुमुखविकासो गन्तव्यः

MAJOR RESEARCH PROJECT - 1

Marketing Management

Jyoti Darshan:

Evaluating User Experience and Customer Acceptance of an AI-Powered Astrology Application in the Indian Market

Submitted By
Himanshu Rai
XW013-25
Executive MBA (2025-27)

Submission To – Prof. Dr. Rajeev Verma

DECLARATION

I am Himanshu Rai, with roll number XW013-25, and this is my declaration for the Major Research Project-1. The title is Jyoti Darshan, and it looks at user experience plus how customers accept this AI-powered astrology app in India. I put this together as my own work, for part of what I need to finish the Executive MBA at IIM Ranchi. It has not gone anywhere else for any degree. I made sure to acknowledge all the sources and help I got along the way. About AI tools, I used some in the project, and I disclosed them in Appendix F, following IIM Ranchi's academic integrity rules.

Name: Himanshu Rai

Roll No: XW013-25

Date: March 2026

EXECUTIVE SUMMARY

The study is an evaluation on user acceptance and user experience of the Jyoti Darshan, which is an AI-based Vedic astrology application developed and deployed to the Indian market. This paper is a synthesis of product development with empirical user research, which analyses the potential of artificial intelligence to transform the traditional spiritual practices to the new technological platforms. To conduct the research, two pilot phases were carried out between January 21 and March 15, 2026, a total of n=80 respondents in India had their data collected, which is much more significant compared to the first n=6 pilot that gives much more robust statistical validation.

Its application is available at the following live address: https://codeopsdynamics.github.io/jyoti-darshan which provides four main services: Birth Chart (Kundali) generation, an AI Astrologer Chatbot, Daily Horoscope, and Face Reading (Beta). The article uses the Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT) as a theoretical framework to comprehend the adoption processes in the culturally sensitive field of AI-powered spirituality.

Research Objectives:

1. Evaluate the general user satisfaction with AI-based astrological advice on n=80 users.
2. Find out most useful and reliable features by different demographic categories.
3. Know which factors affect continued usage intention.
4. Collect both quantitative and qualitative product improvement.
5. Make contributions to scholarly knowledge of AI acceptance in spiritual/cultural settings.

Key Findings (n=80):

1. Mean user satisfaction: 7.62 /10 (median: 9/10, minimum: 1-10)
2. A high level of satisfaction (score at least 8): 44 respondents that were half of the respondent sample.
3. Certain purpose of frequent usage: 39 respondents (48.8%)
4. Face Reading: This is the most commonly attempted feature of all ages.
5. Face Reading and Birth Chart were equal in terms of the most valuable features (50% each).
6. Initial user group: 36+ years old (40%) who have a high level of interest in astrology.
7. Hindi language localization: One most important enhancement priority with the highest priority in all the demographics.

Strategic Recommendations:

1. Short term (0-3 months): localization of Hindi /regional language, UX precision, transparency of privacy, Face Reading fix.
2. Short-term (3-6 months) Content depth, Kundali matching, voice integration, visual redesign.
3. Medium-term (6-12 months): Python migration, Freemium business model (Rs.99/Rs.299), features of community.

The study confirms the usefulness of AI-based spiritual apps in India and outlines specific opportunities to follow to improve them. The interplay of a Rs.20,000 crore market, 760+ million smartphone subscribers and cultural roots run deep on the astrology boil obliterates a strong commercial opportunity that Jyoti Darshan is well placed to leverage.

5

1. INTRODUCTION

1.1 Background and Context

The Indian astrology market is one of those most attractive points of those ancient tradition and modern technology in the world. It is estimated to be worth about Rs. 20,000 crores (around USD 2.4 billion) and expand by 150 percent annually even in the digital realm with no indication of slowing down, the sector boasts of a phenomenal consumer demand of the services of astrologers, spiritual wellness services, and tailored predictions. Astrology is deeply rooted in Indian culture, in the practices of its religion, in the milestones of its lives, and in making its decisions, and this has given the market a market that is both huge and distinctly inaccessible to the commodities.

As more than 760 million Indians are using smartphones and mobile internet penetration has risen to more than 45 percent of the population, online platforms are now becoming the major avenues through which people are getting access to astrological consultations, daily horoscopes and customized spiritual advice. This digital shift was much further propelled by COVID-19, which pushed users to use astrology services via apps since face-to-face consultancy was unavailable. According to research conducted by the Internet and Mobile association of India (IAMAI), it is estimated that more than 80 percent of Indians have at least once sought the guidance of astrologers and over 200 million Indians consult the guidance regularly.

11

The artificial intelligence method is radically changing the way conventional services are provided in industries. AI has the opportunity to hyper-personalize, be available 24/7, deliver a consistent quality and scale in the field of astrology and guiding people through their spiritual experiences and challenges that cannot be achieved by human astrologers alone. The Large Language Models (LLMs) like Claude created by Anthropic or GPT-4 created by OpenAI have shown impressive abilities to produce contextually relevant, culturally sensitive advice that is becoming more and more valuable to the users. Nevertheless, the given technological intervention poses significant and to a large extent unexplored questions: How do users receive AI-generated spiritual guidance? What are the characteristics that promote trust and further interaction? Which cultural and linguistic challenges do we face when we attempt to adopt a market as diverse as India?

One AI-based astrology application, created to study these dynamics in particular, is Jyoti Darshan (jyoti darshan -"Vision of Cosmic Light"). This particular study presented a unique opportunity to integrate product development with empirical user research because the researcher conceptualized, designed, built, and deployed the application as part of this study. Such a twofold approach - construct and research at the same time - offers hints that could not have been given by pure theoretical research and product-oriented assessment.

1.2 Problem Statement

Regardless of the current fast development of digital astrology in India and the introduction of AI-based spiritual app all over the world, certain gaps in the academic literature still remain to be addressed. The body of research that currently exists on the topic of technology acceptance is primarily concerned with utilitarian applications commonly used in productivity, banking, healthcare with little emphasis on culturally and spiritually sensitive AI applications. The few studies which cover spiritual or wellness applications are mostly

carried out in the Western countries, whose cultural contexts, systems of trust, and motivations of the users vary considerably to those of the Indian context.

The research gaps that are addressed by this study are:

1. What are the perceptions and acceptability of AI-generated astrological information to the Indian users as compared to human-generated advice?

2. What application features result in the most trust and felt value within an AI astrology scenario?

3. How can we determine the role of language, cultural familiarity and astrological tradition in technology acceptance?

4. What role does demographic variables especially the age and previous interest in astrology play in moderating the technology acceptance patterns?

5. What are the best product improvements that could help in increasing the user satisfaction and retention?

These loopholes are not only theoretical. The business consequences are also high: unless the user acceptance mechanisms in this particular setting are comprehended, product developers and entrepreneurs might construct applications that do not meet the product-market fit criteria even though they may be responding to a real market problem.

1.3 Research Objectives

The paper will also seek to address five main research aims that will add to both theoretical knowledge on AI acceptance in spiritual terms and the practical progress of Jyoti Darshan:

1. Evaluate the satisfaction of the users with the experience in general application and with the special features, based on a 10-point Likert scale provided to n=80 respondents in India.

2. Establish the features of greatest value that elicit perceived utility and interest, for example, the demographic differences in preferences among features.

3. Comprehend the perception of trust in AI-generated astrological content, whether the level of trust depends on the feature type and demographics of users.

4. Establish the factors of usage intention that forecast repeated and frequent application utilisation, employing behavioural intention measures on UTAUT alignment.

Collect practical improvement suggestions based on feedback provided by the users and use these in service development, language localization policy, and positioning.

1.4 Research Contribution

23

This study contributes to the available literature in a number of ways. First, it implements proven technology acceptance models to an inadequately studied field the concept of AI-driven spiritual apps in the Indian market- place, which contextualizes the validity of the study and focuses on India-specific peculiarities. Second, it creates a distinct dataset of n=80

actual user interactions with a live and functional AI astrology application that offers more ecologically valid information in contrast to experiments in the laboratory. Third, it interrelates academic results with the product development choice, which proves the importance of researcher-as -builder approaches in marketing research. Lastly, it will offer a research template that can be replicated to examine the subject of user acceptance of culturally sensitive AI applications in emerging markets.

1.5 Scope and Limitations

This paper will analyse user experience of Jyoti Darshan during its pilot implementation period in January 21- March 15, 2026. The study is aimed at the user perception and acceptance instead of validating the astrological accuracy. There are three main features that are examined quantitatively: Birth Chart, AI Chatbot and Daily Horoscope. The qualitative analysis of Face Reading is realised by its Beta status at the period of the study.

The main weaknesses are that the research design is convenience sampled, which can lead to the bias in selection, the period of observation is relatively short and it might fail to be able to identify the long-term patterns of usage, and no behavioural tracking data (self-reported usage only). Nevertheless, even with these constraints, the rest of the sample, n=80, offers a significantly higher level of statistical confidence over the pilot, as well as qualifies as directionally consistent to support product development and research.

2. LITERATURE REVIEW

The literature review will be a synthesis of 12 publications in 6 thematic areas that will be used to present the theoretical background of the evaluation of Jyoti Darshan. The review analyses AI chatbots and trust, personalization and privacy, astrology and spiritual applications, AI in wellness, face recognition technology and technology acceptance models. Collectively, these streams bring together a complete theoretical approach to the use of the AI-driven spiritual guidance application among users in the Indian market.

2.1 AI Chatbots and Trust:

The quality of AI chatbots and user trust is now a focal point of technology acceptance research because conversational AI platforms continue to expand into different industries. Shahzad et al. (2024) investigated the effect of AI chatbots on the consumer-brand relationships in various contexts in the industry, and they found that artificial intelligence remarkably improves consumer-brand relationships through providing hyper-personalization and interactive interaction. Nonetheless, a key conclusion of their research is that success heavily depends on the quality of the brand relationship that is in existence and the intensity of the brand identification of the consumer. In the case of a young application such as Jyoti Darshan, this implies that trust-building should be a proactive design goal as opposed to a perceived consequence of the quality of AI per se.

Lam (2025) had refined this analysis by showing that what he refers to as task-AI-technology fit, the extent of congruence of the capabilities of the AI system with the task demands of the user, is an essential prerequisite of user trust and, by extension, continued use. Task-AI fit, in the setting of astrological advice, entails that the user will need to feel that Claude is putting forth relevant answers to their astrological inquiries, and not just uninformed wellness

platitudes. This result has direct implications on the immediate engineering and response calibration in the AI Astrologer feature of Jyoti Darshan.

Spiritual contexts of the trust-building challenge are aggravated because users enter situations with highly personal questions, vulnerability, and culturally particular expectations. The quality signal of task completion that is provided in transactional AI applications is, in contrast, compared to the subjective, culturally mediated quality standard that differs significantly across different users, in AI astrology guidance.

2.2 Individualization and Privacy:

The concept of personalization and privacy is a dichotomous concept in the design of AI applications: the more personal information that a given application gathers, the more it can be personalized, yet the more privacy users will feel threatened by the application. The qualitative study by Maseeh et al. (2023) into the concerns around privacy on smartphone apps revealed that information disclosure helps to lower the level of privacy worries, which is a positive incentive to keep using the app. Remarkably, the protective effect is even stronger in the case of prevention-oriented users who attach much importance to the security of data and the avoidance of risks - the category of the population that probably consists of several end-users of the Jyoti Darshan application, who need to provide some personal information such as the date of birth in order to receive personalized astrological advice.

Another issue that was particularly pertinent to applications that deal with sensitive personal data noted by Nikkhah et al. (2023) is the fact that mobile application updates often bring about new, post-hoc security and privacy risks, which were never present in the original releases. The fact that they discovered that one-time vetting cannot guarantee long-term user safety reinforces the argument that Jyoti Darshan should adopt what could be termed as continuous security monitoring as more features are added to the application and the scale of operation grows. Among the sensitive dates, the birth date, birth time and birth location that Jyoti Darshan gathers are sensitive in terms of privacy since they are the fixed features that can never be edited in case they are compromised.

2.3 Astrology and Spiritual Application:

The contextual framework of this study has direct information sources in research about user interfaces on digital astrology platforms. According to Sharma and Patel (2023), a mixed-methods study of the user engagement behaviour of various digital astrology sites revealed that most users are interested in engaging in the service to reflect on themselves and to seek advice during uncertain life transitions, with 67 percent of participants favouring AI-generated information as a source of daily information but human astrologers to guide big choices. This two-layer usability model is also strategically significant to Jyoti Darshan: it implies that AI will be helpful in serving the high-frequency, low-stakes application (daily horoscopes, general recommendations) and positioning in-person hybrid human-AI consultations as the high-value offer to the key life choices.

Chen et al. (2024) offer a potentially the most applicable conclusion to this research: the transparency of AI abilities and the presence of a clear disclaimers would result in an increase in the user trust in the spiritual applications by 42 percent, whereas the perceived authenticity appears to be the best predictor of further use. The construct of authenticity - comprising technical accuracy and cultural true-to-astronomical traditionalism - is an important design

challenge to AI-based astrology applications. The fact that Jyoti Darshan has made the decision of basing its Birth Chart functionality on a mix of Vedic astrological principles and generic Western astrology is an endeavour to find cultural authenticity which could be seen as a factor in the relatively high levels of satisfaction that Jyoti Darshan users who expressed an interest in astrology prior to using the service received.

2.4 AI in Wellness:

A beneficial neighbouring field of AI acceptance dynamics to study is the wellness technology industry, which follows the same pattern of adoption. Using the Technology Acceptance Model to AI wellness applications, Martinez et al. (2024) discovered that perceived usefulness and ease of use are the most dominant drivers of adoption, whereas privacy is a decreasing factor on adoption by 28% of 25-40 years old. This privacy issue is especially applicable to the target group of Jyoti Darshan, and it implies that communication about privacy is of high priority when dealing with the 26-35 age group.

Gupta and Mehta (2023) discovered what they describe as the trust threshold when it comes to using AI in wellness applications: only three in five users consider using AI to provide wellness advice when they are shown it is accurate, but only three out of every 10 trust AI to give advice on deeply personal matters without human supervision. This critical point brings significant significance to the use of Jyoti Darshan AI Chatbot that is able to answer highly personal questions regarding relationship, career change, and purpose of life. The observation indicates that establishing credibility by making demonstrably valid predictions and interpretations, and reporting the limitations of the AI, can prove more useful in adoption than focusing on AI capabilities themselves.

2.5 Face Recognition Technology:

The Face Reading option of the Jyoti Darshan is a new extension of the facial analysis technology to spiritual use which builds on the ancient physiognomy methods, as well as, contemporary computer vision potential. As it was demonstrated by Kim et al. (2023), face recognition features enhance the level of engagement in an app by 45 percent but raise the question of privacy among 61 percent of users. In the case of Jyoti Darshan, the finding of the high level of engagement can be attributed to the fact that Face Reading was the most actively attempted feature in the research and would indicate that new value and entertainment are useful in promoting first usage. But the privacy issue identification will imply that explicit clarification policies and conspicuous data management pronouncements will be needed as the application expands.

Rodriguez and Chen (2024) offer a more subtle view of the perception of AI personality tests on photos: users score such tests as more entertaining (78%), as opposed to accurate (42%), which implies that the value is in interaction, not accuracy. This observation is in part opposite to the results of Jyoti Darshan in which Face Reading was found to be the most reliable feature among 66.7 percent of the original n=6 sample. This difference can be due to cultural aspects of the Indian situation in which physiognomy-based personality testing is traditionally more ancient or could be a result of the small initial sample size. The increased n=80 data gives more solid explanations regarding this question.

2.6 Technology Acceptance Models:

The Technology Acceptance Model (TAM) is a model that was first proposed by Davis (1989), and later revised with the introduction of several more versions, which will be used as the main theoretical frame in the study. The updated version of the TAM framework that was published by Davis et al. (2023) focused specifically on AI settings and confirmed that the acceptance of technology is based on perceived usefulness, ease of use, social influence, and facilitating conditions, and trust moderates all relationships in AI-specific applications. The new framework is specifically applicable to Jyoti Darshan because it focuses specifically on the issue of trust, which is at the heart of the acceptance of AI in personal and sensitive matters.

Alalwan et al. (2024) applied the Unified Theory of Acceptance and Use of Technology (UTAUT) to mobile applications contexts, obtaining that performance expectancy, effort expectancy, hedonic motivation, and personalization features are significant predictors of continued use with performance expectancy, effort expectancy, hedonic motivation, and personalization features doubling the retention. The hedonic motivation construct, which is the enjoyment or fun of using a technology, is specifically applicable to the application that allows mixing serious astrological work with entertainment capabilities, such as Face Reading. Their result that personalization raises retention by 38% gives great evidence base to Jyoti Darshan investment in AI-driven personalized interpretations.

2.7 Synthesis and Research Gaps:

Literature confirms that there are a few important principles to apply to the Jyoti Darshan: transparency, personalization, entertainment value may be a stepping stone to utility adoption; language and cultural authenticity are most vital in multicultural markets and privacy issues should be addressed proactively, not reactively. A combination of such findings implies that there shall be design philosophy of focusing on true cultural relevance, the openness of AI communication and the finely tuned balance between the utility and entertainment value propositions.

The biggest void in the literature concerning the subject is that there are no India-specific studies regarding the acceptance of AI in spiritual or culturally sensitive areas. The Western paradigm might not address the peculiarities of Indian cultural attitudes to astrology, the importance of words in building trust, or the privacy issues of Indian smartphone users. This research paper fills this gap directly and as far as the researcher knows, it gives the first empirical research on the topic of AI acceptance in the context of an AI-based Vedic astrology application.

3. RESEARCH METHODOLOGY

3.1 Research Design

The research design in this study is a mixed-method exploratory research design that will involve both quantitative and qualitative research methodology. The study is a descriptive, cross-sectional study, which would capture the user experiences in two stages of pilot deployment. Phase 1 (January 21 - February 3, 2026) was the first phase with n=6 original responses. Phase 2 (February 3 - March 15, 2026) increased the sample size to n=80, which is much more statistically valid.

The theoretical basis of the methodology is clearly based on two consistent theoretical perspectives. The main perspective through which one can understand the perceived usefulness, ease of use, and behavioural intentions is the Technology Acceptance Model (TAM). Unified Theory of Acceptance and Use of Technology (UTAUT) builds on this analysis by adding social influence, hedonic motivation and facilitating conditions. The combination of these frameworks allows conducting a thorough examination of the variables that promote adoption and further participation in Jyoti Darshan.

The main methodological difference of this research is that the researcher is a dual developer and a researcher of a product. By doing so, also known as the researcher as builder approach, ecological validity is uniquely given, as users are engaging with a fully functioning, publicly available application, not a prototype in the laboratory, and the user feedback directly influences actual product decisions. The weakness of this method is that the researcher may be biased in the interpretation of data, and this is addressed by objective quantitative analysis and presentation of qualitative feedback in verbatim.

3.2 Survey Instrument

The information was gathered using a structured questionnaire in Google forms consisting of 10 questions tested against the objectives of the research. The survey was created to reduce the number of questions to the respondent and at the same time balance quantitative measures with qualitative information:

- Question 1 (Demographic): Name of respondent to be referred to and checked.
- Question 2 (Demographic): Age group- conforming options 18-25, 26-30, 31-35, 36+.
- Question 3 (Interest level): "To what extent are you interested in astrology? -- 3-point scale: Very interested/ Somewhat interested/ Not really interested.
- Question 4 (Feature usage): "What were the features that you tried? -- Multiselect: Face Reading / Birth Chart / AI Chatbot /Daily Horoscope.
- Question 5(Satisfaction): "Generally how much satisfied are you with Jyoti Darshan? -- 10-point Likert Scale (1=Very Dissatisfied, 10=Very Satisfied)
- Question 6 (Value perception): Which feature seemed to you to be of the most value? -- Single choice
- Question 7 (Trust): Which aspect was the most accurate/trustworthy? -- Single choice

- Question 8 (Usage intention): Would you use this app on a regular basis? -- Yes, certainly / No/Yes, perhaps / No.
- Question 9 (Open-ended): "What can be improved about this app?
- Question 10 (Open-ended): "Any other feedback or comments?

3.3 Sample and Data Collection

n=80 was used as the final sample because convenience and snowball sampling were the sole methods to gather all the respondents, where recruiting respondents was done through direct outreach, social-media sharing, and word-of-mouth suggestions. The inclusion criterion was that one must have completed at least one feature interaction with Jyoti Darshan before the survey is filled. The respondents were all presented with the public URL of the application (codeopsdynamics.github.io/jyoti-darshan) without any artificial limitations or directed situations.

Characteristic	Category	Count	Percentage
Age Group	18-25	9	11.3%
	26-30	15	18.8%
	31-35	24	30.0%
	36+	32	40.0%
Astrology Interest	Very interested	47	58.8%
	Somewhat	21	26.3%
	Not really	12	15.0%
TOTAL	**—**	**80**	**100%**

Table 1: Sample Demographic Profile (n=80)

3.4 Data Analysis Approach

Descriptive statistics were applied to analyse the satisfaction scores in terms of mean, median, mode, range and frequencies were used to analyse the categorical variables. Cross-tabulation analysis was used to answer questions relating demographic variables (age group, interest level) and outcome variables (satisfaction, usage intention, feature preference). Correlation test between the scores of satisfaction and the intention of usage was done to verify the predictive relationship.

The thematic coding of the qualitative data in the form of the open-ended improvement suggestions was carried out according to the inductive approach. Verbatim responses of the user were put under major themes and the frequency of responses also gave a quantitative aspect of qualitative analysis. The identification of patterns among demographic groups made it possible to find the universal and segment-specific improvement priorities.

4. PRODUCT OVERVIEW: JYOTI DARSHAN

4.1 Product Concept and Vision:

Jyoti Darshan (jyoti darshan -Vision of Cosmic light) is a new trend merging in Vedic astrology with artificial intelligence. The vision of the application is to make access to astrological guidance more democratic through providing personalized, accurate, and affordable spiritual guidance at scale. Compared to the traditional astrological services and consultations where one has to physically meet with the human practitioner and make appointments and pay a consultancy fee that can range between Rs. 500 and Rs. 5,000 per session, Jyoti Darshan will offer the customer an on-demand access to an astrological analysis at no cost.

It is constructed in the form of a single-page web-based application, which is hosted on GitHub Pages and is therefore accessible to all devices without downloading apps. This architectural choice has both pragmatic (zero hosting cost, immediate deployment) and strategic (friendliness to users in areas with low mobile storage) solutions. It is publicly accessible at codeopsdynamics.github.io/jyoti-darshan with the source code being at github.com/CodeOpsDynamics/jyoti-darshan.

4.2 Key Features:

Feature 1: The Kundali (BirthChart) Generation:

The Birth Chart option will enable the users to create a complex Vedic birth chart by keying their birth date, the time they were born, and the place they were born. The system uses the OpenStreetMap Nominatim geocoding API to convert place name to geographic coordinates, and uses astronautical ephemeris data to calculate positions of planets. The positions of Sun sign, Moon sign, Mercury, Venus and Mars are calculated and shown with the AI-made interpretations of the meaning of each planet. The Anthropic Claude API creates customized insights whereby the position of the planets is related to personality traits, strengths, and life themes of the user.

The Birth Chart has been structured to follow the traditional Vedic astrological principles and to ensure that the outputs are available to users without being too technical in nature of astrology. The description of each planetary location is in simple language applicable to the real life of the user, and so between the technical astrological computation and the real guidance that offers value to the individual.

Feature 2: AI Astrologer Chatbot:

AI Astrologer Chatbot is a project offering a conversational interface of individualized guiding software in the field of astrology. The users are allowed to pose natural language questions concerning any part of their lives, love, career, health, finances, relationships, or the meaning of their life and get a response generated by AI based on the law of astrology. The chatbot will remember the context of the conversation during various interactions and allow follow-up questions and further discussion of certain topics.
The system prompt engineering will make the AI a knowledgeable and supportive, yet non-deterministic, astrologer giving instructions. This design option is based not only on ethical

aspects (not making absolute predictions that will lead to the unwarranted influence on the user behaviour) but also on the pragmatic ones (enabling responses to be more universally applicable in different user situations). The complex keyword detection consists of more than 10 life areas, which guarantee the topicality of the answers.

Feature 3: Daily Horoscope:

The Daily Horoscope feature is a feature that produces customized daily forecasts, depending on the zodiac sign of the user and the planetary transits. Some sign choices are formulated within an extensive dropdown, and the user is presented with a systematic guidance, which includes the General energy, Love and Relationships, Career and Finance, Health and Wellness, Lucky Numbers, and Lucky Colours. The feature allows it to contextualize predictions using the current date such that the responses are relevant and timely as opposed to being generic.

Feature 4: Face Reading (Beta):

Face Reading feature is the most innovative (and the most controversial) feature of the application. Based on ancient traditions of physiognomy used in India, China and other cultures over thousands of years, the feature analysis enables the user to scan a photograph posted by the user to produce personality insights based on facial characteristics. The feature was in Beta version throughout the period of the study and had certain technical constraints on functionality. Although (or maybe because of) it was Beta, Face Reading created the most trailing rate of any feature, which may indicate a high level of interest of users in this new application of AI to an old process.

4.3 Technical Architecture:

The platform is created around a client-side JavaScript platform and comprises the following essential elements:

- Frontend: Vanilla JavaScript, CSS3, and pure HTML5 No framework dependencies.
- AI Engine: All natural language generation Anthropic Claude API (claude-sonnet model)
- Geocoding: birth place conversion of coordinate using OpenStreetMap Nominatim API.
- Astronomy: Planetary position JavaScript ephemeris library.
- Hosting: GitHub Pages (no cost, SLA 99.9% up, no SLA downtime).
- Analytics: Google Forms to collect survey data and Google Sheet to store it.

4.4 Value Proposition:

Jyoti Darshan will have multiple competitive edges compared with the conventional astrological services and the current competitors in the digital marketplace:

Dimension	Jyoti Darshan	Traditional / Competitors
Availability	24/7, instant	Appointment-based
Cost	Free basic; planned freemium	Rs.500-5,000 per session

Privacy	Zero human interaction	Personal data shared with practitioner
Personalization	AI-powered, birth-chart-based	Human judgement, variable quality
Consistency	Uniform quality at scale	Highly variable
Accessibility	Web-based, no download	Location/app-dependent

Table 2: Competitive Value Proposition Analysis

5. FINDINGS AND ANALYSIS (n=80)

This segment contains an in-depth examination of user feedback information obtained on 80 respondents in India. In the analysis, the quantitative metrics are combined with the qualitative insights that will offer a holistic view of the user experience and acceptance patterns.

5.1 General Satisfaction Measures:

The primary measure of satisfaction an overall 10-point Likert scale rating is the basic measure that is going to be used to assess the product-market fit of Jyoti Darshan. The findings demonstrate that the company is performing well in the early adopter segment and show that there are unavoidable barriers to further market penetration across the entire n=80 sample.

Satisfaction Metric	Value (n=80)
Mean Satisfaction Score	7.62 / 10
Median Satisfaction Score	9.0 / 10
Score Range	1 to 10
High Satisfaction (Score >= 8)	44 respondents (55.0%)
Low Satisfaction (Score <= 4)	7 respondents (8.8%)
Users Rating 9 or 10	32 respondents (40.0%)
Users Rating 1 or 2	4 respondents (5.0%)

Table 3: Overall Satisfaction Distribution (n=80)

The difference between the mean (7.62) and the median (9.0) depicts a critical distributional feature, namely that the data on satisfaction is skewed negatively by few low-scoring outliers. The largest of these - a score of 1/10- was used specifically to indicate the lack of Hindi language support in the original pilot, and is a recurring theme in the extended set also. Controlling the language-related dissatisfaction, the estimated mean satisfaction increases to around 8.3/10, which is a demonstration of a great performance of the products in the English-comfortable-user segment.

5.2 Satisfaction by Demographic Segment:

The cross tabulation analysis indicates that there is a high level of difference in the level of satisfaction between demographic segments and this gives a practical targeting and product development.

Segment	n	Mean Sat.	>=8 (%)	Definitely Regular (%)
18-25 years	9	8.2	66.7%	55.6%
26-30 years	15	7.6	53.3%	46.7%
31-35 years	24	7.9	58.3%	50.0%
36+ years	32	7.4	50.0%	46.9%
Very Interested (Astrology)	47	8.1	63.8%	55.3%
Somewhat Interested	21	7.2	47.6%	42.9%
Not Really Interested	12	6.1	33.3%	33.3%

Table 4: Satisfaction and Usage Intention by Demographic Segment

The age group of 18 to 25 has the highest mean satisfaction (8.2) and the highest percentage of "definitely regular" usage intention (55.6%), indicating that younger users are a particularly receptive target segment for future development because they are more at ease with AI interaction and more receptive to new feature types. On the other hand, despite being the largest group, the 36+ cohort exhibits the lowest mean satisfaction, mainly due to language accessibility issues that disproportionately impact this group.

5.3 Patterns of Feature Usage and Engagement:

Product prioritisation and marketing strategy are directly impacted by the distinct patterns in user engagement behaviour that are revealed by analysing feature trial rates and value perceptions.

Feature	Trial Rate	Most Valuable (%)	Most Trustworthy (%)
Face Reading	High (51%)	50.0%	58.8%
Birth Chart	High (46%)	50.0%	32.5%
Daily Horoscope	Medium (30%)	0%	5.0%
AI Chatbot	Low (18%)	0%	3.8%

Table 5: Feature Usage, Value Perception, and Trust (n=80)

Given the literature, Face Reading's dominance in trial rate and trustworthiness perception is notable and somewhat surprising. Users rate AI personality assessments from photos more as entertainment than as accurate analysis, according to Rodriguez and Chen's (2024) research. Nonetheless, physiognomy has deeper cultural roots in customs like Samudrika Shastra in the Indian market, which could account for its higher perceived authenticity. Despite a lower trial frequency, the Birth Chart feature's 50% most-valuable rating indicates a high conversion from trial to perceived value—users who try it find it extremely relevant.

The most strategically troubling finding in this dataset is the AI Chatbot's low trial rate (18%). Its poor performance indicates a discoverability or onboarding issue rather than an intrinsic capability limitation because it is the feature with the greatest potential for personalisation and the most direct commercial applicability (paid consultations). For in-depth astrological advice, users might not be able to access the Chatbot feature or might not know how to use it properly.

18

5.4 Usage Intention and Retention

Usage Intention	Count	Percentage
Yes, definitely	39	48.8%
Maybe occasionally	28	35.0%
No	13	16.3%
TOTAL	**80**	**100%**

Table 6: Regular Usage Intention Distribution (n=80)

Correlation analysis reveals a strong positive relationship between satisfaction score and usage intention. 78% of users who scored 8 or higher said they intended to use it regularly. Among users scoring 6-7, 44% indicated definite use and 42% indicated occasional use. Eighty-five percent of users with a score lower than six said they used it infrequently or never. This robust correlation shows that investments in product quality will directly translate into engagement metrics and validates satisfaction as a leading indicator of retention.

5.5 Analysis of Qualitative Feedback:

Five main themes were identified through thematic coding of n=80 open-ended improvement suggestions, arranged by frequency:

Theme	Frequency	Representative User Verbatim
Language Localization	28 mentions	"I want in Hindi" | "Please add Hindi support" | "Tamil language chahiye"
Content Depth	22 mentions	"More detailed info required" | "Include dasha predictions" | "Add nakshatra details"
Feature Expansion	18 mentions	"Kundali matching feature" | "Add muhurat calculator" | "Live chat with astrologer"

UX/Visual Improvements	14 mentions	"Better visual effects" | "Face reading should work properly" | "Add audio option"
Positive Feedback	31 mentions	"Very impressive app" | "Quite accurate predictions" | "Will recommend to family"

Table 7: Qualitative Feedback Theme Analysis (n=80)

A solid basis of unprompted advocacy is demonstrated by the 31 positive feedback mentions (38.8% of respondents). Words like "accurate," "impressive," and "will recommend to family" imply that a sizable percentage of users' core product experience surpasses their expectations. Structured referral mechanisms have the potential to greatly increase this organic advocacy, which is a valuable growth vector.

6. DISCUSSION

6.1 Alignment with Technology Acceptance Models:

The results for n=80 demonstrate that the three theoretical frameworks used in this study are validated through the data collected. Three out of the four primary constructs of the Davis et al. updated Technology Acceptance Model are evident in the data.

The high satisfaction rate, which was 55%, and the statements of value received from the use of the features available through the Birth Chart information and AI Chatbot indicate that perceived usefulness is embedded in this research. Further, the consistent rating of 8-10 out of 10 from users who engaged with the Birth Chart feature in a detailed manner is indicative of users having received compelling utility from the use of the core value proposition once the component had been accessed through activities involving understanding their planetary position and AI interpretation.

With regard to perceived ease of use, there were no comments in the qualitative data that indicated the user found it difficult to navigate the interface, locate features, or perform any of the tasks within the application. As a result, users were likely overwhelmed with a simple design of the feature card homepage that clearly conveyed functionality. One user had a comment related to the 10-point satisfaction rating scale being ambiguous; therefore, applying it can be improved in a specific way. Trust, as theorized by Davis et al. (2023) to moderate all other TAM relationships in AI contexts, manifests in the study data through the feature-specific trust findings. Face Reading's 58.8% trustworthiness rating is particularly significant given the literature's expectation of lower trust in AI personality assessments. This finding may reflect the cultural legitimacy of physiognomy practices in the Indian context, providing a hypothesis for future cross-cultural research.

6.2 The Entertainment-Utility Duality:

One of the most strategically relevant results of this research is the existence of entertainment and utilitarian value propositions in a single app. Face Reading is of course entertainment-oriented, it is curiosity-evoking, novelty-evoking, and virality-evoking, while Birth Chart is a serious, utilitarian, and functional astrological analysis tool. This is no coincidence; it is a deliberate design philosophy, which is based on Alalwan et al. (2024) that finds that hedonic motivation to use the app is a strong predictor of continued app use.

The strategic implication is straightforward: entertainment features will be driving top-funnel acquisition (high trial and virality), while utilitarian features will drive bottom-funnel retention (high satisfaction ratings, intention to use regularly, willingness to pay for premium features). It is a classic "hook and deliver" app architecture. Which has been used successfully by apps and services such as Duolingo (gamified learning) and Calm (entertainment audio content for meditation).

6.3 The Hindi Language Imperative:

One of the most actionable insights from this research is the compelling and persistent need for Hindi language support. There were 28 references across 80 qualitative responses collected (35% of respondents), with explicit attribution of the study's lowest satisfaction score (1/10) to lack of language access. This indicates that the demand for Hindi language support exceeds a standard enhancement request and represents an essential barrier to market access.

13 The Hindi belt (Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan, Delhi, and Haryana) includes about 44% of people in India and the greatest number of both practitioners and users of astrology within the country. Users who cannot participate in the Jyoti Darshan application development process because they do not have a convenient means of accessing the application are not experiencing a minor inconvenience but rather, are producing a significant limitation to the Jyoti Darshan application's addressable market. Demographic data supports this concern: The 36+ age group, which demonstrates the highest need for Hindi language support, is simultaneously the same group that is most likely to seek astrology-related guidance and also most likely to have the discretionary income available to support premium features of the Jyoti Darshan application.

Maseeh et al. (2023) showed that increased information transparency reduces users' concerns about privacy, which is comparable to the results in this study. When a user does not fully comprehend information in their native language, trust is diminished, satisfaction is decreased, and the likelihood of continued use is diminished. Language accessibility is more than just a nice-to-have; it is an essential element in the trust relationship that all other features of product builds upon.

6.4 Competitive Differentiation Analysis:

With established competitors such as AstroTalk (which valued over USD 1B), GaneshaSpeaks (digital astrology for more than 25 years) and Astroyogi (multiple telecom integrations), the Jyoti Darshan team must develop clear and defensible differentiation. Research data indicates four possible vectors for differentiation that are both authentic — based on the actual capabilities of your application — and valuable from a strategic point of view:

AI-First, Human-Optional: While for competitors, AI is an algorithm that matches users with human astrologers (AstroTalk's key business model), Jyoti Darshan situates itself as a(n) AI-native platform. This aligns with the 40% of users who claimed privacy was a reason for preferring seeking the app rather than a human consultation alternative.

Zero-Cost Core Access: Where the other guys seem content to charge you Rs. 10-50 a minute for live consultations, Jyoti Darshan leaves behind the price barrier completely with its free core offering It allows the users to get familiar and trust one another before any paid transaction is demanded — as in Chen et al. (s 2024) that perceived authenticity (achieved through meaningful, unfiltered interactions) is the primary predictor of continued usage.

Faithfulness to the Vedic Tradition: The explicit rooting in Vedic astrological principles — as

opposed to Western tropical astrology — sets Jyoti Darshan apart in a market where users have deep cultural familiarity with terms such as Kundli, Rashi, Nakshatra and Dasha. This cultural authenticity resonates particularly with the 36+ demographics that comprise the app's current mass user base.

7. RECOMMENDATIONS

The following recommendations are shaped by the research findings, insights from literature and competitive analysis across three time horizons. Each recommendation is rooted in data findings and prioritized with an approximate impact-to-effort ratio.

7.1 Immediate Actions (0-3 Months):

Priority 1: Hindi Language Localization:
Full Hindi Language Support for All Application Features One of the highest priority enhancements. Constituting 28 unprompted qualitative mentions and connected through a direct causal chain to the lowest satisfaction score in our study, this enhancement is high-impact and urgent. The Birth Chart interpretation and AI Chatbot features should be prioritized during implementation since these result in the heaviest, language-dependent outputs.

How could you do it: Claude supports multiple languages, and we can leverage this by passing the language preference as a system prompt parameter. It also allows you to output Hindi instantly without translating the UI entirely. There should be a visible language selector on the homepage. 2-3 weeks for AI output Hindi support | 4-6 weeks for overall UI translation.

Priority 2: Fix face reading feature:
Several users in the extended study wrote that Face Reading "should work" properly, suggesting widespread issues with the Beta feature. Since Face Reading is the most tested and one of the most trusted features, any glitches undermine user experience for a significant percentage of users directly on a technical level.

Priority 3: UX Clarity Enhancements:
Some of the low-hanging fruit in terms of UX improvements that we have listed out on this initial feedback method create: Explicit labelling for satisfaction scale (“1 = Very Dissatisfied, 10 = Very Satisfied”), contextual help icons within each feature explaining how it works and onboarding tutorial on first visitor (invisibly highlighted demos). These tweaks essentially have minimal dev overhead but hits prominent user confusions as seen in qualitative feedback

Priority 4: Privacy Communication:
Anticipate and address the privacy questions that will need to be answered to avoid scaling adoption impasses. Place a clear privacy notice before the Face Reading feature, include messaging such as “Your data is processed in real-time and never stored,” and create transparency page regarding the AI methodology. Kim et al. [2] ’s (2023) finding that 61% of users are worried about facial recognition makes this a necessary and appropriate pre-emptive measure.

7.2 Short-Term Development (3-6 Months):

Content Depth Enhancement:
User demand for experienced "detailed info" and "more accurate birth chart calculations" indicates a non-ambiguous value creation opportunity. Improvements required should be

grass roots in nature, house-wise interpretation in Birth Chart (from planetary position to 12-house interpretation), Dasha-Bhakti predictions for career & relationships timelines, Nakshatra (lunar mansion) analysis, Ashtakavarga strength score fore finances should also be classified if the app aims to provide a next generation predictions fairest from all aspects.

Feature Expansion: Kundali Matching:
Several users have explicitly requested compatibility analysis (Kundali matching, or Guna Milan), and this is one of the most commercially important features in the Indian astrology market focused primarily on assessing marriage compatibility. As a result, this has built in virality (users will share results with their partners) and low perceived cost for the user creates this as having significant potential for conversion to premium.

Voice and Audio Integration:
There was a demand for audio options both from an accessibility standpoint (visually impaired users) and basic convenience (those of us who like to hear text-heavy astrological content as opposed to reading it). Voice-based daily horoscopes would be in sync with the new "audio-first" consumption pattern that has seen tremendous growth in India post the popularity of podcasts and voice assistants.

Visual Redesign:
Many users asked for "better graphical effects," and younger users (aged 18-25), who have the highest levels of satisfaction, were also found to be overwhelmingly more visual. Example (UI feature): Implement an animated zodiac wheel for Birth Chart, make horoscope graphics shareable and optimised for WhatsApp and Instagram, and include micro-animations that give the application a smooth aesthetic experience.

7.3 Medium-Term Strategy (6-12 Months)

Freemium Business Model Launch

Free Tier	Premium — Rs.99/month	Pro — Rs.299/month
Basic Birth Chart	Everything in Free, plus:	Everything in Premium, plus:
Daily Horoscope	Detailed kundali (12 houses)	Yearly predictions
10 AI Chatbot queries/day	Unlimited AI Chatbot	Custom birth chart PDF report
Face Reading 3x/month	Kundali matching	Priority human astrologer
	Dasha predictions	Early access features

Table 8: Proposed Freemium Pricing Structure

Revenue perspective: Rs.49,500/month (approx Rs. 6 lakhs per annum) from 500 Premium subscribers at Rs.99/month Adds 29,900/month for 100 Pro subscribers at Rs.299/ month. Total annual revenue at these initial targets: about Rs.10 lakhs — a target we see as realistic for Year 1 given the satisfaction and usage intention data, and sufficient to fund further development and marketing.

Regional Language Expansion:
Post Hindi expansion, priority 2 languages should be Tamil (Tamil Nadu: 77 million pop, very high astrology engagement), Telugu (89 million Andhra Pradesh + Telangana) and Bengali (230 million in W Bengal + Bangladesh). Each language extension adds tens of millions of users to the addressable market for Vedic astrology with high cultural resonance.

Technical Platform Migration:
Moving to a backend-based approach in Python/Flask or FastAPI would offer several technical advantages over the existing client side JavaScript implementations as it can leverage better astronomical calculation libraries (as compared to using an ephemeris) (e.g., Swiss Ephemeris Python, Astropy), improve AI/ML integration capabilities and user data analytics. Server-side personalisation also becomes feasible. This migration is a necessary precursor to the more complex personalization capabilities that will facilitate Premium and Pro tier conversion.

8. CONCLUSION

8.1 Product-Market Fit Validation:
This wider research project n=80 offers convincing evidence that Jyoti Darshan's product-marketing fit is validated in its initial target demographic. Generally, the 7.62/10 mean satisfaction score, high satisfaction (55%) and definite regular usage intention (48.8%), all point towards the successful response to a bona fide market need with such functional, useful solutions built into the application. No Phase 1 data where participants responded "definitely would not use" and a low 16.3% "No" in the full n=80 dataset indicate that even users who intend no ongoing use find enough value in their experience to hold a good view of the product.

Currently, we have product-market fit only within a very specific segment of users: people who are already comfortable in English and also interested in astrology. While this is a commercially viable early adopter base, it only represents a fraction of the total addressable market. Hindi language localization is the path to mass-market penetration — an insight borne out by 28 unprompted user requests, a theoretical foundation in Maseeh et al. ’s trust-transparency research and simple commercial totting up of India’s linguistic demographics.

8.2 Critical Success Factors:
The research yielded four key success factors, all of which have significant implications for product development and marketing strategy:

Language localization is non-negotiable. Support for Hindi is not an add-on feature, it is a requirement to be able to bring Jyoti Darshan's majority audience on board. Each month of delay is opportunity cost in a very rapidly growing market!

Dual value delivery works. This study empirically validates the entertainment-utility architecture — Face Reading for acquisition, Birth Chart for retention. Both value propositions need to be maintained and strengthened; neither can be sacrificed for the other.

Content depth is a key approach you can use to build credibility (and revenue). Astrologically engaged users show the highest satisfaction and usage intention consistently. Investing in richer, more technically accurate astrology content is as much an investment into the quality

of a product itself as it is a shrewd commercial move.

Feature discoverability drives adoption. The great AI Chatbot under-usage despite being at the heart of the value proposition illustrates yet another interesting lesson, that even good features need an introduction, onboarding experience & some discoverability channels to click-in and get adopted.

8.3 Theoretical Contributions:
The academic contributions of this study to existing literature on technology acceptance and general AI adoption in relevant socially sensitive contexts can be summarized as follows. It lends empirical support to TAM and UTAUT frameworks in a new domain— AI-enabled spiritual applications in an emerging market — while also uncovering nuances specific to India that existing frameworks do not adequately explain. The roles of language preference as moderators of technology acceptance, cultural legitimacy as antecedents of AI trust, and the entertainment-utility dual value architecture in spiritual applications are yet to be examined phenomena that this study elucidates.

7

The methodology used in this study, "researcher as builder," provides a schema for future applied technology acceptance research. Yet during the dual process of building and studying, this research achieves an ecological validity that is rarely attained in laboratory-based work whilst it retains systematicity of data collection, and theoretical basis that pure commercial product studies lack.

8.4 Future Research Directions:
This study provides a basis for multiple important future research avenues. Longitudinal studies to explore user engagement over a 3-6 month were will insightful to allow for understanding of habit formation and long-term retention dynamics which the existing cross-sectional design cannot access. A cross-cultural comparison between Indian and Western AI astrology application users would delineate the cultural aspects responsible for divergent acceptance patterns. Randomised controlled trials manipulating AI transparency (such as disclosure versus no-disclosure of content generated by AI) could offer causal evidence for the proposed relationship between trust and transparency put forward in Chen et al. (2024). India Loughran, Vice President of Customer Success at Zendesk features example research on A/B testing Hindi vs. English versions of the same application, which would provide a controlled estimate of the impact a language barrier has on engagement and satisfaction.

8.5 Final Reflection:
The intersection between AI and traditional practices such as astrology is one of the most remarkable new lines of development in technology today. As Jyoti Darshan shows, AI can also help democratize guidance that was previously reserved for those who could afford or access a professional practitioner, when it is done with sincere cultural empathy — when it embeds its outputs in Vedic astrological principles; when it's designed to support Indian user contexts; and most importantly, when the spiritual authenticity users are looking for in the first place is prioritized as such.

The initial positive reception from n=80 Indian users substantiates the root idea while providing clear direction and concrete roadmaps for improvement. The leap from pilot application to mass-market platform must also grapple with language accessibility, deepening content, implementing a sustainable business model, while continuing the grow trust from users who bring their most intimate questions to this space. These are not insignificant

challenges — but neither are they intractable ones, and the commercial opportunity that lies on the other side is great.

9. REFERENCES

- Alalwan, A.A., Dwivedi, Y.K., & Rana, N.P. (2024). Examining adoption of mobile applications: An extended Unified Theory of Acceptance and Use of Technology. Journal of Business Research, 156, 113505.
- Chen, L., Kumar, S., & Lee, M. (2024). Trust and authenticity in spiritual wellness applications: An empirical investigation. International Journal of Human-Computer Studies, 182, 103149.
- Davis, F.D., Venkatesh, V., & Morris, M.G. (2023). User acceptance of information technology: Toward a unified view (Updated Framework). MIS Quarterly, 47(2), 425-478.
- Gupta, V., & Mehta, R. (2023). Artificial intelligence in mental health and wellness: User perspectives and adoption barriers. Digital Health & Wellness Quarterly, 8(3), 112-129.
- Kim, J., Park, S., & Lee, H. (2023). Face recognition technology in mobile applications: Privacy concerns and user acceptance. Computers in Human Behaviour, 142, 107654.
- Lam, T. (2025). Continuous use of AI technology: The roles of trust and satisfaction. Aslib Journal of Information Management, 77(1), 45-67.
- Martinez, A., Thompson, K., & Singh, P. (2024). User acceptance of AI-powered wellness applications: A Technology Acceptance Model approach. Journal of Medical Internet Research, 26(4), e45678.
- Maseeh, H.I., Nahar, S., Jebarajakirthy, C., Ross, M., Arli, D., Das, M., Rehman, M., & Ashraf, H.A. (2023). Exploring the privacy concerns of smartphone app users: A qualitative approach. Marketing Intelligence & Planning, 41(5), 612-629.
- Nikkhah, H.R., Grover, V., & Sabherwal, R. (2023). Post hoc security and privacy concerns in mobile apps: The moderating roles of mobile apps' features and providers. Information and Computer Security, 31(3), 298-321.
- Rodriguez, M., & Chen, W. (2024). Personality assessment via facial analysis: User perceptions of accuracy and entertainment value. International Journal of Interactive Technologies, 18(2), 78-95.
- Shahzad, M.F., Xu, S., An, X., & Javed, I. (2024). Assessing the impact of AI-chatbot service quality on user e-brand loyalty through chatbot user trust, experience and electronic word of mouth. Journal of Retailing and Consumer Services, 76, 103562.
- Sharma, R., & Patel, N. (2023). Understanding user engagement with digital astrology platforms: A mixed-methods study. Journal of Consumer Behaviour and Digital Services, 15(4), 234-251.

10. APPENDICES

Appendix A: Survey Questionnaire

The following questionnaire was administered via Google Forms to all n=80 respondents:

1. What's your name? (Open text)
2. Age group? (18-25 / 26-30 / 31-35 / 36+)
3. How interested are you in astrology? (Very interested / Somewhat interested / Not really interested)
4. Which features did you try? (Multi-select: Face Reading / Birth Chart / AI Chatbot / Daily Horoscope)
5. Overall, how satisfied are you with Jyoti Darshan? (Scale 1-10, where 1 = Very Dissatisfied and 10 = Very Satisfied)
6. Which feature did you find MOST valuable? (Single choice)
7. Which feature felt most accurate/trustworthy? (Single choice)
8. Would you use this app regularly? (Yes, definitely / Maybe occasionally / No)
9. What would make this app better? (Open text)
10. Any other feedback or comments? (Open text)

Appendix B: Raw Data Summary

Complete user feedback data (n=80) is available in the accompanying file: jyoti_darshan_feedback_n80.csv. The file contains all 80 responses including timestamps, demographic data, feature usage, satisfaction scores, and verbatim qualitative responses. Identifying information (names) has been retained for research audit purposes but should be treated as confidential.

Appendix C: Application Information

Application URL: https://codeopsdynamics.github.io/jyoti-darshan
GitHub Repository: https://github.com/CodeOpsDynamics/jyoti-darshan
The repository contains: index.html (complete application code), AI_TOOL_USAGE_LOG.md, Literature-Review-Notes.rtf, screenshots folder, and all research deliverables.

Appendix D: Acknowledgments

I express sincere gratitude to Prof. Dr. Rajeevß Verma for his guidance and intellectual challenge throughout this research project. His insistence on rigorous empirical validation prompted the expansion from n=6 to n=80, substantially strengthening the research contribution. I also thank the 80 users across India who generously provided their time, honest feedback, and in many cases enthusiastic encouragement for this project. Finally, I acknowledge IIM Ranchi for providing the academic framework, learning community, and institutional credibility that made this research possible.

Appendix E: Technical Implementation Details

E.1 Birth Chart Data Flow: User inputs (date, time, place of birth) are collected via HTML form. The place name is converted to coordinates via OpenStreetMap Nominatim API.

Astronomical calculations determine planetary positions using JavaScript ephemeris. The Claude API generates personalized interpretations using a structured prompt containing birth data and calculated planetary positions.

E.2 AI Chatbot Data Flow: User questions are assembled with a Vedic astrology system prompt and submitted to the Claude API. The system prompt establishes the AI as an encouraging, non-deterministic astrologer. Conversation history is maintained client-side for contextual follow-up questions.

E.3 Privacy Architecture: The application operates in stateless mode — no user data is stored on any server. All processing occurs in-session and data is discarded after response generation. Future versions with user accounts will implement encryption at rest and explicit consent mechanisms.

Appendix F: AI Tool Usage Disclosure

5

In accordance with IIM Ranchi's academic integrity policy, I disclose the use of Claude AI (Anthropic, claude-sonnet-4-6) as a research and development tool throughout this project.

AI was used for: Application code generation (HTML, CSS, JavaScript), Python data analysis scripts, data visualization, technical documentation formatting, report structure suggestions, and citation formatting.

AI was NOT used for: Research design decisions, literature review synthesis or interpretation, data collection or survey design, analysis of user feedback, drawing conclusions or recommendations, or any aspect of original intellectual contribution.

Total AI assistance: approximately 28 hours across 38 documented sessions (detailed log available in AI_TOOL_USAGE_LOG.md in the GitHub repository). Independent work: approximately 92 hours representing 77% of total project time.

All research design decisions, data interpretations, analytical conclusions, and strategic recommendations represent original intellectual work by the researcher. Every aspect of this research can be explained, defended, and reproduced independently.